

Manual

Power Recording in Energy Management EGM-LEE 8.1

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1 Power Recording in Energy Management

Purpose

Integrated into the Energy Management module, the performance values which power meters provide are recorded, documented and monitored for limit value infringements with this package.

You use the function package when:

- You wish to record and save energy-related performance values from the meters.

Integration

The EMG-MGM basic function is a precondition for the power recording.

Separate evaluation modules are available for online visualization and graphic evaluation.

Features

Application service (AS) with functions for recording and processing of energy-related performance values and other process parameters such as active power, flow rates, temperatures, etc.:

- Functions for creating and editing of master data for performance values and special parameters (e.g. action and tolerance limits) for monitoring the created performance values
- Direct recording of the performance values of power meters and storage of the measured value curve at defined intervals
- Generation of performance values and further indicators on the basis of the values recorded by physical and virtual power meters
- Online limit value monitoring during the performance value recording
- Storage and logging of limit value infringements
- Triggering of alarms at limit value infringements in conjunction with the HYDRA escalation management, incl. control of signal transmitters or possibility of further processing of alarms in external systems (e.g. machine and system controllers)
- Tabular evaluation of the limit value infringements on the basis of the defined monitoring parameters
- Performance log for individual measuring points over selected periods
-

2 Logical channels

Overview

Menu	Master data → Process data processing → Logical channels
Transaction code	lgchcnf
Function authorization	lgchcnf.*

The logical channels form the central point where logical configurations, physical machines and shop floor servers (terminal type PCC) come together.

Purpose

You must configure the logical channels in order to assign the process parameters/PDV events:

- to machines
- to shop floor servers (PCC)
- to physical channel numbers.

The logical channels define which technical channel is used on a terminal to record the characteristics.

Logical channels specify the "channel mapping".

You can also use this configuration to specify the physical channels in more detail, for example:

- you can specify how data is collected
- you can define outputs e.g. for alerts or
- you can define target values.

As of PDV 8.2, you can also configure the channels for the recording of TAGs (e.g. serial number, etc.).

This way, you can connect the measured values to an alphanumeric TAG and you can perform analyses based on TAGs in defined evaluations.

Logical channels are versioned master data. For each data record, a validity period is therefore specified (exact to the second). Only in the period specified, the data record is valid for the system.

Note: A modification of the channel configuration is not passed on to the terminal, i.e. the new channel configuration is made available to the terminal only after the terminal/ shop floor components are restarted or after the cyclical start of the configuration monitor service (e.g. in certain situations such as target value modifications).

Integration

The logical channels are used to assign process parameters or PDV events to a machine-shop floor server combination.

Selection criteria

The application provides the following selection criteria:

"General" tab

Channel

Selection of channel number

Designation

Selection of channel designation including wildcard function and search screen.

Channel type

Selection of channel type using selection box

Data class of the channel

Selection of data class using selection box

"Assignment" tab

Machine

Selection of machines including wildcard function and search screen

PCC terminal

A selection of terminal/shop floor server is possible. A search screen may also be used.

"Activation/alert" tab

Active

Selection of active channels only

Alert

Selection of logical channels with enabled option *Alert*

Valid from-to

Start and end time including date and time to specify the validity period

Alert channel

Selection of the logical channels where the respective alert channel is stored.

Field descriptions

Tab "General"

Channel

Channel number on the machine. This number must be unique for the PCC terminals. Only values between 1 and 9999 are permitted.

Designation

Optional, logical designation of the logical channel

Channel type

Includes the reference to the data that is collected via the channel. This can either be a configured PDV event (E) or a process parameter (PP, corresponds to the decimal value or the alphanumeric tag) of a characteristic.

Data class of the channel

For channels of type PP, the data class of the channel defines which value of a process parameter is collected using the channel. The values supported are:

- MV: Measured value/ alphanumeric tag
- TV: Target value
- UTL: Upper tolerance limit
- UPAL: Upper process action limit
- LPAL: Lower process action limit
- LTL: Lower tolerance limit



If a target value or a tolerance or process action limit is passed via a channel, it must be guaranteed that a time span of at least 30 minutes is respected between two value changes of a channel.

Data type (available as of PDV 8.2)

Data type that is recorded via the channel. The following three options are available:

- Decimal
- Alphanumeric
- Tag (alphanumeric)

The data types *decimal* and *alphanumeric* are used for the typical process parameters. The tag values recorded are available as selection criteria in the selection criteria based on tags. These values can be selected.

The permitted characters for the data types are defined as follows:

- Decimal: Numeric decimal values
- Tag (alphanumeric): 0-9 a-z A-Z äöüÄÖÜ _-+#,.,

Note:

The recording of tags is limited to 50 characters. If the data source transfers a tag with more than 50 characters (alphanumeric), the shop floor client cuts off the values after 50 characters.

Tab "Assignment"**Machine**

Assignment of the machine

PCC terminal

Assignment of the PCC terminal

Process parameter

Assignment of a process parameter if the channel type is defined for the collection of a process parameter.

Event

Assignment of a defined event if the channel type is defined for the collection of an event.

Tab "Properties"**Input type**

The input type of the channel is used to define how the shop floor server is to access the machine data. The control parameters are used for the collection of process values and of events (see parameter type). The values supported are:

- A: Automatic (if supported by driver)
- T: Trigger controlled
- C: Cyclic

Cycle time

If the channel is controlled using a cycle time, this field contains the number of seconds after which the values are queried.

Trigger

If the channel is controlled via trigger channel, the channel number of the trigger is stored in this field. Only values between 1 and 9999 are permitted.

Direction

Direction of data flow for the channel. Values can be transmitted from the machine to the data collection (input, column value I). This is the direction normally used to record machine data. In addition, the data direction can be configured from the data collection to the machine (output, column value O) to overwrite a value depending on the selected data class of the channel. For example, setting a new target value for the connected machine.

Tab "Activation/alert"**Active**

The logical channel is switched to the active status here. In data collection, only active and valid channels are used. The activation or deactivation of a channel is only valid after the restart of the respective shop floor components.

Valid from-to

Validity period during which the channel is used for data collection.

Alert

This option specifies if a configured alert channel is activated (output signal for setting a physical signal). The activation or deactivation of a channel is only valid after the restart of the respective shop floor components.

Alert channel

This option specifies the physical channel number where an output signal is set when an alert condition is fulfilled. Depending on the channel type, the alert can be triggered by a PDV event or a limit violation of a process parameter. Only values between 1 and 999 are permitted.

Configuration of an alert channel

In order to activate an output signal in case of a limit value violation of a process parameter, an output channel of the type "Process parameter" must be created.

The following settings are made in the dialog "Insert logical channel".

- For the channel number, you configure a "virtual channel" with any value (between 1 and 9999). However, this number must be unique. This channel number is used to manage the combination "Alert channel – data class of the channel".
- You use the "Data class of the channel" to define the limit value violation that triggers the alert.
- You must also specify the process parameter that is monitored.
- Activate "Output" as direction.
- Enable the options "Channel active" and "Alert active".
- The alert channel must match the physical channel where an output signal is set in case of an alert condition.

For a detailed description of individual input fields, refer to section "Field description".

Checking Business Parameter Containers (BSCs)

See [here](#) for further information on checking the system with respect to business parameters.

3 PDV Characteristics Master Data

Overview

Menu	Master data → Process data processing → PDV characteristics master data
Transaction code	chrp
Function authorization	chrq.*

You use the catalog of characteristics to predefine characteristics that are then used in collection rules. It aims at persons involved in data collection planning. The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options.

PDV characteristics master data

Main page Profile

Request data Cancel Print preview Print all Insert Copy Edit Delete Save Help on operation Help on application Help on detail application Help

Selection

Main page User fields

Characteristic no.

Characteristic designation

Process parameter

PDV characteristics master data

Drag a column here to group by that column

Characteristics	Charact...	Process pa...	Formula	Interval type	Interval va...	Unit
AirHumidity	Air Humidity	AirHumidity		None		
BeltVelocity	Belt Velocity	BeltVelocity		None		
CycleT	Cycle Time	CycleT		None		
CylinderTemp	Cylinder Te...	CylinderTemp		None		
DosingT	Dosing Time	DosingT		None		
Inject.T	Injection Time	Inject.T		None		
InnerPress	Inner Press...	InnerPress		None		
InnerTemp	Inner Temp...	InnerTemp		None		
MoldTemp1	Mold Tempe...	MoldTemp1		None		
MoldTemp2	Mold Tempe...	MoldTemp2		None		
NozzleTemp	Nozzle Tem...	NozzleTemp		None		
OuterTemp	Outer Temp...	OuterTemp		None		
PartsWeight	Parts Weight	PartsWeight		None		

PDV characteristics master data

Characteristics Specifications Storage Visualization Inspection - computation Administration User fields

Characteristic

Characteristic no.

Characteristic designation

Process parameter

Formula

Purpose

The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options. Only enter data in the characteristics catalog that need not be changed when later assigned to a collection rule. Do not define limit values, for example. This is normally not useful because the limit values are only known when the collection rule is created. Only when you create and assign a collection rule, a relation to a concrete article or machine is established. Note this and you will know what kind of information you should predefine. For example, it must be carefully considered whether the characteristic "outer diameter" is only created once and detailed information is stored in the collection rule later on or whether several "outer diameter characteristics" are created, e. g. with specification of limit values. Usually, it is an advantage to store a small number of general characteristics. The required evaluations/reports also play a role in this context. If a new "outer diameter characteristic" is created for almost every tolerance change, this characteristic is "valid" for one article or machine only. In a subsequent failure analysis, a comprehensive evaluation is not possible in this case!

It is important that you can still change any definition stored in the catalog or add missing definitions when you later plan the collection rule.

Important

The configurations made in the characteristics' master data need not be definitive for the collection rule. The characteristics' master data is used as a template when the collection rules are later created. You can complete and change all configurations of the characteristics' master data in the collection rule.

Integration

The characteristics' master data is used as a template when the collection rules are later created. The field *Process parameter* provides a logical connection to the logical channels. The connection is only established when the characteristics are definitively defined in the collection rules.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
Characteristic number
- Designation:
Name of the characteristic
- Process parameter:
Process parameter

If several selection criteria are used, the overlapping results are displayed below in the *Characteristics master data*.

Field descriptions

Find below a description of the columns and input options for characteristics:

Tab *Characteristics*

Characteristic no.

Unique number of the characteristic

Characteristic designation

Name of the characteristic

Process parameter

Predefinition of the process parameter

Formula

See section "Formula calculation" below.

Tab *Specifications*

Select the *Specifications* tab to enter the sampling scheme and the tolerance limits. Note: As mentioned above, it is only reasonable to define tolerance limits in the characteristics master data under certain circumstances.

Sampling scheme (no longer relevant as of PDV 8.3)

The following sampling schemes are available:

- None
- Piece-related

With a piece-related sampling scheme, you can define the interval. All single values collected in this interval are then combined and form one sample. If you define a piece-related sampling scheme, you can make evaluations for samples, for example control charts.

Interval value (no longer relevant as of PDV 8.3)

Interval used to combine the single values collected in this interval to a sample.

Display of failures in the Failure Mode Analysis (PDV 8.1 and PDV 8.2)

To display the created failures in the Failure Mode Analysis, you must specify an interval value. Process parameters without specified interval value are not integrated in the Failure Mode Analysis.

**Field visibility**

The field is only visible and input is only possible if the user is assigned the function authorization "InspectionInterval".

Unit

Pieces, meter, kg, etc. Units are assigned using the unit catalog.

Decimal places (no longer relevant as of PDV 8.1)

Enter the decimal places. Leading zeros before the comma are not displayed in the specification fields. By default, the number of decimal places defined in the system settings is pre-assigned.

Size (measure type)

Validation and tolerance limits can be entered as absolute, relative or percentage values. Note: You must enter relative or percentage limits as negative values.

Upper PL

Specifies the upper process limit. In PDV, this value also defines the displayed upper red area in process visualization.

Upper TL

Specifies the upper tolerance limit (upper specification limit)

Target value

Specifies the target value

Lower TL

Specifies the lower tolerance limit (lower specification limit)

Lower PL

Specifies the lower process limit. In PDV, this value also defines the displayed lower red area in process visualization.

Upper TL – Generate failure / Lower TL

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:TG> or AUTO:TG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Specifications process**Upper PAL**

Upper process action limit

Lower PAL

Lower process action limit

Generate failure

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:PEG> or AUTO:PEG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Storage (no longer relevant as of PDV 8.3)

Using the filter functions for storage, you define the frequency used to store single values. This value affects the further processing of measured values on the HYDRA server. In later evaluations, only the actually stored values can be displayed. The storage frequency is also the basis for further aggregations by the PDV Distributor calculating samples. Here, the PDV Distributor only accesses data of the online data set.

Example: The defined filter specifies that only one of ten measured values is saved. At the same time, a sample interval specifies that 5 measured values are combined to one sample. After filtering, the storage frequency is the basis for the calculation of samples. 5 measured values are then actually combined to one sample. Result: For 50 measured values, which have actually been collected, one sample is generated.

Filter function:

Define a filter function for the collection of PDV values. The following four filter functions are available: *None*, *Cyclic*, *Frequency* and *Percentage*. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- *None*

No filter, the system saves each collected value.

- *Frequency*

Enter a number of measured values. This means that the system only saves every n measured value. The system evaluates the measured values collected in the meantime but they are not saved.

- *Cyclic*

Enter a time interval in seconds. This means that the system only saves a measured value every n seconds. The system evaluates the measured values collected in the meantime but they are not saved.

- *Percentage*

Enter a time interval and a percentage. In the PDV data collection, the system saves the

measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is saved because of the percentage deviation. The system evaluates the measured values collected in the meantime but they are not saved.

Visualization

Visualization

Check this option to enable the online visualization for the process parameter.

Position

Use this integer field to specify the visualization position of the process parameter. Use this for graphics in the online visualization with more than one display element on one page. Example: If 16 display elements are shown on one page, you can use this option to define the position of the different process parameters.

If the value 0 or less is specified, the process parameter is not visualized. You can only edit the input field after you have checked the option *Visualize*.

Visualization *Filter function*

Here, the same filter functions are available as for the storage, but you can use this second setting to decouple the storage from the online visualization. For example, you can configure that each measured value is stored, but only every tenth measured value is visualized.

The following four filter functions are available: *None*, *Cyclic*, *Frequency* and *Percentage*. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- *None*
No filter, the system visualizes each collected value.
- *Frequency*
Enter a number of measured values. This means that the system only visualizes every n measured value.
- *Cyclic*
Enter a time interval in seconds. This means that the system only visualizes a measured value every n seconds.
- *Percentage*
Enter a time interval and a percentage. In the PDV data collection, the system visualizes a measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is visualized because of the percentage deviation.

Inspection – computation

Check characteristic (no longer relevant as of PDV 8.3)

You can use this option to define whether the process parameter is checked against limit value violations in the PDV data collection or whether the collected measured values only pass the data collection to be stored.

This field also affects the online visualization. You can only display characteristics that are processed by the logic of the PDV data collection and are not only stored.

Formula parameters only

Use this option to define that the created process parameter is only used as parameter to calculate a further process parameter. If you set this option for a process parameter, this process parameter will never assess measured values with respect to violated limit values and these process parameters will never be stored in the database.

Compute limit values

If you enable this option, the fields for the calculation of target values or limits are activated. Formula calculation, see section "Formula calculation"

Upper TL formula

You can use the formula for the calculation of the UTL to calculate a new UTL in combination with other parameters. If you recalculate the UTL, this will also lead to changed target values. See section 3.5 for further information on formulas.

Upper PAL formula

You can use the formula for the calculation of the UPAL to calculate a new UPAL in combination with other parameters. If you recalculate the UPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Target value formula

You can use the formula for the calculation of the target value to calculate a new target value in combination with other parameters. If you recalculate the target value, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower PAL formula

You can use the formula for the calculation of the LPAL to calculate a new LPAL in combination with other parameters. If you recalculate the LPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower TL formula

You can use the formula for the calculation of the LTL to calculate a new LTL in combination with other parameters. If you recalculate the LTL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Formula calculation

Using the formula calculation function, you can calculate the measured values collected via machine connection before you evaluate them. For example, you can add an offset to the calculation of a characteristic, you can directly convert units in the collection process or you can perform calculations in combination with other characteristics.



To store and display a formula, the user must have the function authorization `iriscp.formula`. If the user is not authorized, the field is not displayed.

You specify the calculation of measured values in the application *Characteristics* in tab *General*. You can also calculate the target values of an automatically collected characteristic and the upper/lower tolerance or action limits.

Important: HYDRA versions below MES Weaver 2.0 with server systems AIX, HP-UNIX, SCO-UNIX or DEC ALPHA do not provide this kind of formula calculation.

The first part of the formula specifies the level where the formula calculation is made. The following types are available:

- V - Calculation on the level of single values without self-reference.
For each measured value of the characteristics involved, exactly one single value is generated for the calculated characteristic.
- O - Calculation on the level of samples with self-reference.
Use this type of formula to refer to the value itself and to integrate limit or target value.

After the above identifier, the actual formula is specified. The following operators, functions and constants are supported:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the natural logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the common logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x to the power of y
Constants	
pi	3.141592654
e	2.718281828

If constant numeric values are used in formulas, you must be careful not to use thousand separators. If these constants are floating point numbers, be careful to use a dot as decimal separator instead of a comma.

The following syntax [A:B:C]. applies for the variables that identify the single or default values of the process parameters involved. The available values are listed below.

You can specify the following values for section A:

- X – single value/measured value
- UTL – Upper Tolerance Limit
- UPL – Upper Process Action Limit
- TV – Target Value
- LPL – Lower Process Action Limit
- LTL – Lower Tolerance Limit

Note: The value X defined for section A (single value/measured value) is only used for the calculation of measured values; this means, the value is not used in the target value formulas or formulas of the upper/lower tolerance or action limits.

Note: Aggregate functions like MAX, MIN or AVG, which are used in HYDRA CAQ, are not supported with automatically collected characteristics.

Section B describes how the relevant characteristic is identified. The following possibilities are available:

- SELF – self-reference
(This requires formula type O)
- PPARAM – Reference to further process parameters of the same machine
With this formula, only process parameters of the same machine can be calculated.

Section C identifies the characteristic using the field content specified in section B. This means: If a self-reference is specified in section B, a section C may not exist. If instead a reference to a further process parameter is stored for PPARAM, the process parameter must be specified in section C.

Example 1:

The process parameter always collects ten times the rounded value from the machine connection.

→ Formula: O: `round([X:SELF]) * 10`

Example 2:

The characteristic "area" results from the product of the process parameters LENGTH (LAENGE) and WIDTH (BREITE). For each single value of the two source characteristics, a single value is calculated for the characteristic "area".

→ Formula: V: [X:PPARAM:LAENGE] * [X:PPARAM:BREITE]

Example 3:

The following formula is not stored for the measured value, but for the target value of the process parameter "speed". The target value is calculated using the own target value plus the target value of process parameter DURCHMESSER (diameter). The target value for "speed" is therefore recalculated, if the target value for "diameter" changes.

→ Formula: O: [TV:SELF] + [TV:PPARAM:DURCHMESSER]

Note:

The formula calculation is performed directly after reception of the measured values from the PDV data collection. Changes to any of the process parameters included in the formula are then used as events that trigger recalculation.

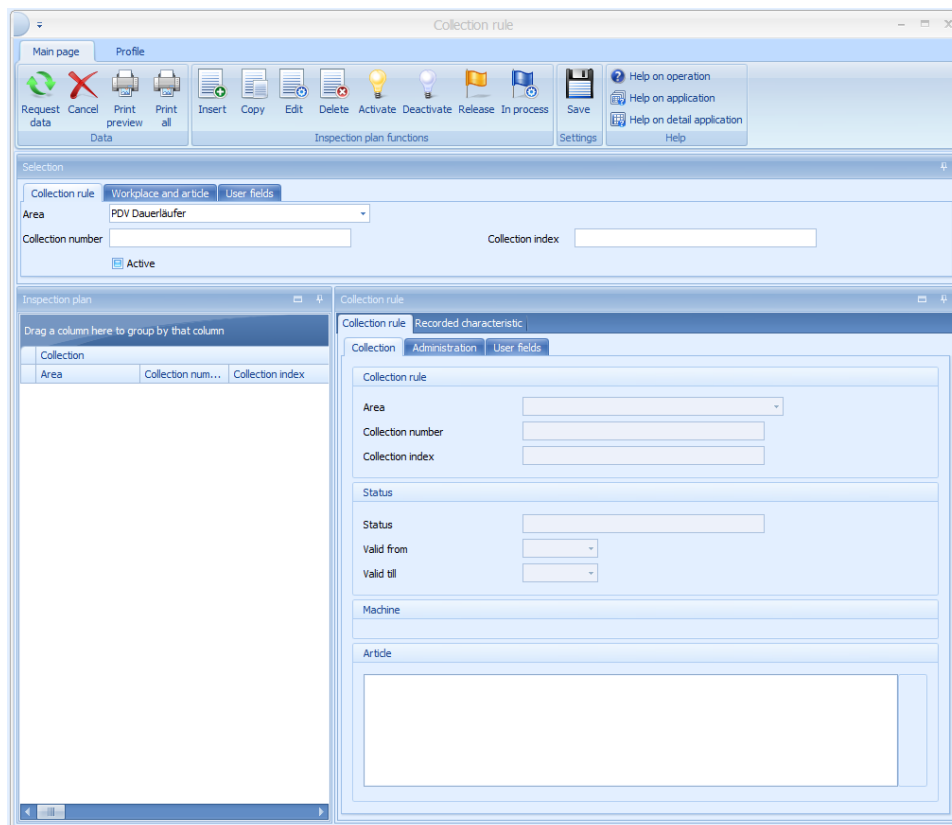
4 Collection Rules

Overview

Menu	Quality management → Process data collection → Collection rules
Transaction code	iplpd
Function authorization	iplpd

Purpose

The automatic data collection of process characteristics is based on the collection rules specified in the Process Data Collection module.



Integration

Collection rules result in collection requests. Collection requests activate the defined rules for data collection. Characteristics are the basis for collection rules. You can take the characteristics from the characteristics master data catalog. The generation of collection requests activates the actual data collection process. The data collection uses the logical channel that is defined by the process parameter of the characteristic.

Requirements

The characteristics catalog must be defined.

Selection criteria

The application provides the following selection criteria:

You can choose from three different tabs to select data. You can use the tabs "Collection rule", "Workplace and article" and "User fields".

Collection rule:

Area (drop-down list)

You can use the drop-down list to select the required area.

Collection number

Select the collection number.

Collection index

Select the collection index.

Active checkbox

Enable the checkbox to select active collection rules.

Workplace and article:

Machine

Select the required machine.

Article number

You can use a search application to search for and select an article number.

Article name/designation

You can select the article name.

User fields:

Object type

You can select an object type.

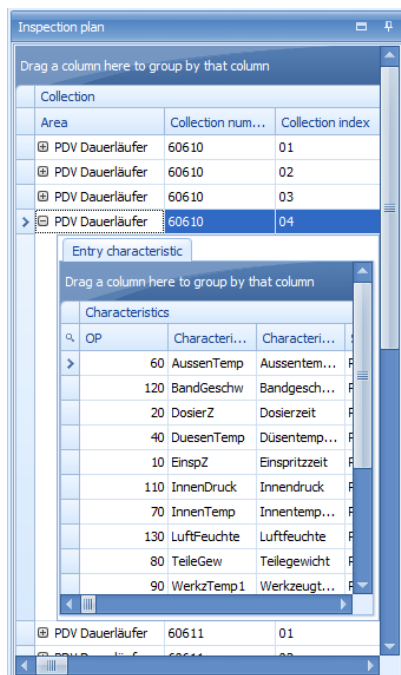
User field key

Use the drop-down list to select a user field key.

Field descriptions

The table view of the detail application offers an overview of existing entries. You can sort the displayed information using table functions. The displayed data complies with the specified selection parameters.

The table view is an integrated master-detail table, which contains not only the collection rules, but can also be expanded to show the recorded characteristics:



Depending on the selected line, the details pane shows the corresponding information. The tabs vary accordingly.

When you call up an editing function, this function affects the entry/entries selected in the table.

Toolbar

In addition to the standard buttons for creating, deleting and editing, the following, context sensitive buttons are available for the collection rules:



Activate

Function authorization: `iplpd.activate`

Click this button to enable the selected collection rule. Only one version of a collection rule can be active at a time! If you activate continuous monitoring rules, the system immediately generates a collection request. This activates data collection. If you activate article-related collection rules, the system automatically generates a collection request the next time you log on a production order for this article.



Deactivate

Function authorization: `iplpd.deactivate`

Click this button to deactivate the selected collection rule. If you deactivate continuous monitoring rules, the system terminates the collection request.



Release

Function authorization: `iplpd.release`

Click this button to release the selected collection rule. You can no longer edit released rules. You can only activate released rules.



In process

Function authorization: `iplpd.inprocess`

Click this button to cancel the release for the selected collection rule. You can now edit the rule.

Detail applications

The detail view shows the information matching the entry selected in the table view.

The detail view consists of two tabs, i.e. the collection rule and the recorded characteristic. These tabs include some additional sub-tabs.

Collection rule:

Collection: The detail view shows the data that has to do with data collection. These include the collection rule, the status, the machine and the article.

Administration: This detail view displays the data that has to do with data management. This is primarily a time stamp for the selected inspection plan.

User fields: Shows the object type and the user field key.

Recorded characteristic:

Characteristics: This detail view shows the characteristics for the current inspection plan.

Specifications: This detail view shows the samples and dimensions in detail. You need the function authorization *iriscp.interval* in order to store an interval value in the sampling scheme.



Please note that sample-related reports/evaluations are not available for process characteristics whose limit values are controlled via specification lists.

Storage: Shows the selected filter function.

Visualization: Detail view for the visualization.

Inspection - computation: Detail view for inspection and calculation.

Administration: This detail view displays the data that has to do with data management. This is primarily a time stamp for the selected inspection plan.

User fields: Shows the object type and the user field key.

5 Collection Request

Summary

Menu	Quality Management → Process Data Collection → Collection Request
Transaction code	crpd
Function authorization	crpd

The collection requests are generated according to defined rules from the collection rules. The actual collection of data is activated by the collection requests.

Usage

The collection requests are generated by

- Continuous collection rules, fully automatic when the collection rule is activated. When it is deactivated, the request is automatically closed again.
- Article related collection rules, fully automatic when production orders for the defined article are generated or logged on.

Integration

The collection requests are created from the collection rules and then transferred to the collection computer in compressed form so that it starts the collection after it receives the rule.

Normally no manual intervention into the collection requests is necessary because they are created and ended automatically. Rather, the function is used only for purposes of analysis.

Requirement

In addition to the collection rules, the collection of the logical channels must be defined so that collection can occur.

Toolbar

The toolbar contains function for controlling the activity of the collection requests such as release, complete and cancel.

6 Violation of Limit Values, tabular

Summary

Menu	Quality management → Process analysis → Tabular violations of limit values
Transaction code	plima
Function authorization	plima

The "tabular violations of limit values" function is used for the process analysis in quality management.

Usage

In this application, the collected limit value violations of the process characteristics are shown. The limits are defined in the collection rule. The table shows the violations and the return to the "OK range" indicating the point in time and the measured value.

Integration

The limits must be defined in the collection rule.

Selection parameters

The following selection criteria are available in the application:

Machine

In the machine pool application, the desired machine can be selected.

Process parameters

In addition, a process parameter can be input.

Event from - to

Restricts the event period by defining the start and end time. In addition, the start and end time may be selected by defining a relative date.

Violation

Restricted by the type of violation using a drop down list

Consider long-term data

Long-term data may also be taken into account by clicking the relevant checkbox.

"Tabular violations of limit values" detail application

This tabular report shows the infringed limit values recorded and saved in the database including the following information:

Machine

Machine where the limit value violation was recorded

Process parameters

Technical name of the characteristic or the entered process parameter

Event timestamp

Time when the limit violation was recorded

Value

Value of the recorded process parameter by which the limit has been exceeded

Violation

Specifies which limit defined within the collection rule has been infringed

7 Process Analysis (based on machines)

Summary

Menu	Quality management → Process analysis → Process analysis (based on machines)
Transaction code	tpdam
Function authorization	tpdam

The process analysis, based on machines, is used for the process analysis in quality management.

Usage

The table view of the detail application offers the user an overview of the existing entries. The display of the information can be sorted using the representation in the grid and is oriented based on the specified selection parameters. The display of the fields can be adjusted for each user with the context menu.

Integration

The function displays collected measured values of process characteristics. These values must have been previously collected in the system.

Selection parameters

The following selection criteria are available in the application:

Machine

A detailed search can be made for the machine using the pool application.

Time domain from - until

Specifies the time interval to be selected.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Tag type + tag ID

Selection of the tag type and tag ID

Consider long-term data

Long-term data is taken into account.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition to this, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support

Detail application: process analysis (based on machines)

This tabular report shows the process characteristics recorded and saved in the database including the following information:

Characteristic

Technical name of the characteristic or the entered process parameter

Designation

Defined characteristic name

Target value

The target value defined for this recorded process parameter at that specific point in time

UTL

The upper tolerance limit defined for this recorded process parameter at that specific point in time

UPAL

The upper process action limit defined for this recorded process parameter at that specific point in time

LTL

The lower tolerance limit defined for this recorded process parameter at that specific point in time

LPAL

The lower process action limit defined for this recorded process parameter at that specific point in time

Unit

Unit of the recorded characteristic

Measured value

Measured value recorded for the characteristic at the time of measurement

Time of measurement

Time of measurement of the characteristic

Detail application: PivotTable

This report shows the process characteristics recorded and saved in the database in a pivot table including a graphic.

